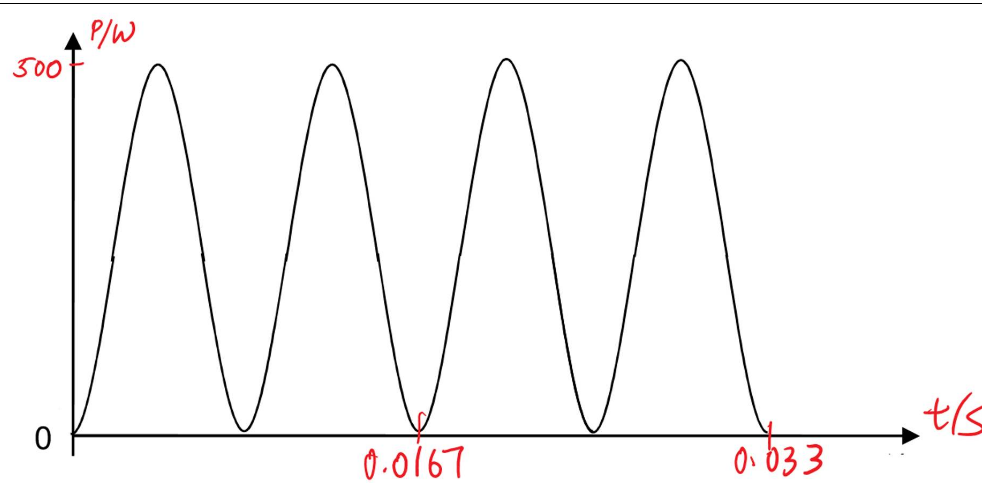
	Hence no opposing force can push the ring upwards.
5(a)	*The root-mean-square (r.m.s.)* value of an alternating current is the *steady direct current* which converts electrical energy to heat in a given resistance at the same rate as the a.c.
5(b)(i)	$V = 170\sin 377t$ and comparing with $V = V_o \sin wt$ $V_{\text{rms}} = \frac{V_o}{\sqrt{2}} = \frac{170}{\sqrt{2}} = 120\text{ V}$
5(b)(ii)	$w = 2\pi f$ $f = \frac{w}{2\pi} = \frac{377}{2\pi}$ $= 60\text{ Hz}$
5(b)(iii)	Mean power, $P = \frac{V_{\text{rms}}^2}{R} = \frac{(120)^2}{58} = 248\text{ W}$
5(c)	 Note: - The gradient of points on the x-axis must be zero. - Draw a line at P=250W. The shape of the curve must be "symmetrical" about this line
6(a)	A *photon* is a quantum of electromagnetic radiation with energy given by $E = hf$